

For Those With DSP Skills

Typically, DSP engineers today write codes to implement a function on a DSP, architect hardware blocks and systems, and design algorithms. In this article, we take a look at the DSP scenario in India, career scope in this field and industry experts' suggestions to aspiring engineers



(Image courtesy: www.vit.ac.in)

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Digital signal processing (DSP) has two facets—hardware design and software algorithms and applications. Traditionally, India has been a power house of software and hence has a strong DSP software presence, believes Abhijit Giri, senior engineering manager, DSP, Analog Devices India. A number of young engineers develop optimised algorithms in India.

Giri says, "In DSP software domain, there are companies which develop either off-the-shelf DSP software IPs such as various multimedia and communication codec modules, or provide system integration services to OEMs."

Talking about hardware, he adds, "Analog Devices have strong DSP design presence coupled with ownership to improve the performance and take

hardware to the next level to enable software engineers to develop better algorithms providing better customer experiences. With combined hardware and software developments, I believe

that DSP is a great domain poised to provide good opportunities for engineers in India."

Explaining why the DSP domain is growing and has a bright future in India, Rajeev Kaushal, general manager-product engineering services, elfnchips, says, "There are more than a hundred organisations dealing with DSP and more than 20 companies specially focused on the DSP domain in India. It is a very niche market. Today, there is a lot of requirement related to DSP in the defense vertical. It is a domain that is growing very well in India."

He feels that in the coming days, there will be lots of projects and career opportunities in DSP domain in India.

Scope

"DSP technology finds a number of applications," informs Kantheti Srinivas, director-product applications and systems engineering, Analog Devices

Suggestions from industry experts

"For a software and systems engineer, key skills needed are sound mathematics, signal processing concepts, digital systems followed by strong programming skills. Hardware designers should have skills related to semiconductor design, architecture focused on improved performance, power optimisation, etc"

—**Kantheti Srinivas, director-product applications and systems engineering, Analog Devices India**

"Those interested in pursuing a career in DSP should have studied at least one DSP processor architecture. In addition, they should be conversant in 'C' language and Assembly language programming, and familiar with software optimisation methodologies. Also, a module on DSP cores is included in electronics engineering courses. In case you have not studied DSP cores, many colleges offer short-courses in embedded systems and DSP programming. It is advisable that you enroll in such a course"

—**Subodh Shukla, Engineering Group Director, Cadence Design Systems (India)**

"Some basic to intermediate-level courses in digital signal processing/processors are offered by Cranes. These courses are authorised by companies like Texas Instruments. There are some courses offered by M.S. Ramaiah School and IISc as well"

—**Sunil Kshatri, AVP, human resources, Sasken**